

Record Full-C band Unrepeated Transmission over 532-km HCF achieving 30.66-Tb/s with ROPA and 15.73-Tb/s without ROPA

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Abstract We report the first full-C-band unrepeated transmission over 532-km of hollow-core fiber using high-power DFA and 135 GBd WDM channels, yielding 15.73 Tb/s throughput. Leveraging low loss of the HCF for both signal propagation and remote pumping, we also achieve 30.66-Tb/s net throughput with ROPA. ©2026 The Author(s)

Introduction

Unrepeated optical links offer a cost-effective solution for connecting remote areas due to their simplicity allowing for reduction of costs and risks of failures. These systems optimize the trade-off between data rate and distance by using high-power transmitters, low-loss fibers, and advanced techniques such as distributed Raman pumping or remote optically pumped amplifiers (ROPAs) [1]. The advent of hollow-core fibers (HCF) marks a significant breakthrough, addressing the limitations of conventional fibers by guiding light through a central air core. HCFs can enable long-distance transmission with reduced losses and high-power levels without nonlinear penalties, thereby minimizing the constraints on amplification techniques and potentially reducing operational costs and maintenance. Recent advances in AR-HCF technology, including the demonstration of low propagation losses [2–4] and wideband transmission capabilities [5–6], have accelerated their adoption as a promising medium for next-generation optical networks.

A recent demonstration has underlined the capabilities of HCF for high-performance unrepeated links [7]. In silica-based fiber, the longest single-carrier unrepeated systems distances have instead been obtained with EDFA (erbium-doped fiber amplifiers) amplification, advanced Raman pumping and remote optically pumped amplifiers (ROPAs) [8]. Record single-wavelength unrepeated transmission over 746.69 km offline using Raman and ROPA amplification in silica fiber [9], and up to 41 Tb/s over 300 km in wavelength division multiplexing (WDM) transmission and 32 Tb/s over 407 km using several Raman and ROPA amplifiers [10–15]. The corresponding single wavelength HCF record reaches 611.9 km with EDFA alone, and 400-Gb/s at 726.1 km [16]. Reaching comparable performance in a WDM system, as

required for commercial deployment, remains highly challenging.

This study explores full C-band WDM unrepeated transmission systems, without and with ROPA over HCF systems. Using only doped fiber amplifier (DFA) we achieve a total net capacity of 15.73-Tb/s over 532 km. The second system with the same reach, achieves a total record net capacity of 30.66-Tb/s, leveraging ROPA amplification. Both transmission systems use Gap-Tube-Assisted Support-Tube Hollow Core Fiber (GTA-ST-HCF) [4] and one 34-dBm high power Erbium-Ytterbium co-doped fiber amplifier (HP-EYDFA). Our experiments demonstrate a significant improvement in distance over previously published WDM results as in Fig. 1.

Experimental setups

Experimental setups for “Setup 1” without ROPA (solid line, Fig.2(a)) and “Setup 2” with ROPA (dashed line, Fig.2(b)) are shown, both of them relying on the GTA-ST-HCF and the HP-EYDFA. GTA-ST-HCF exhibits low-IMI of -68.8 dB/km which has been shown to be advantageous for long-distance transmission [17]. For both setups, the dummy channels are generated from amplified spontaneous emission (ASE) noise sources. A waveshaper (WS) filter spectrally shapes the output of ASE noise and delivers a WDM

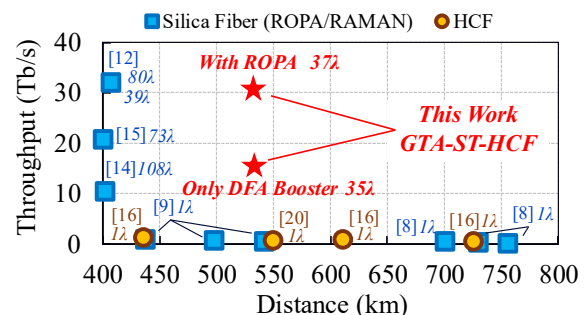


Fig. 1: State-of-the-art unrepeated transmission in C-Band.

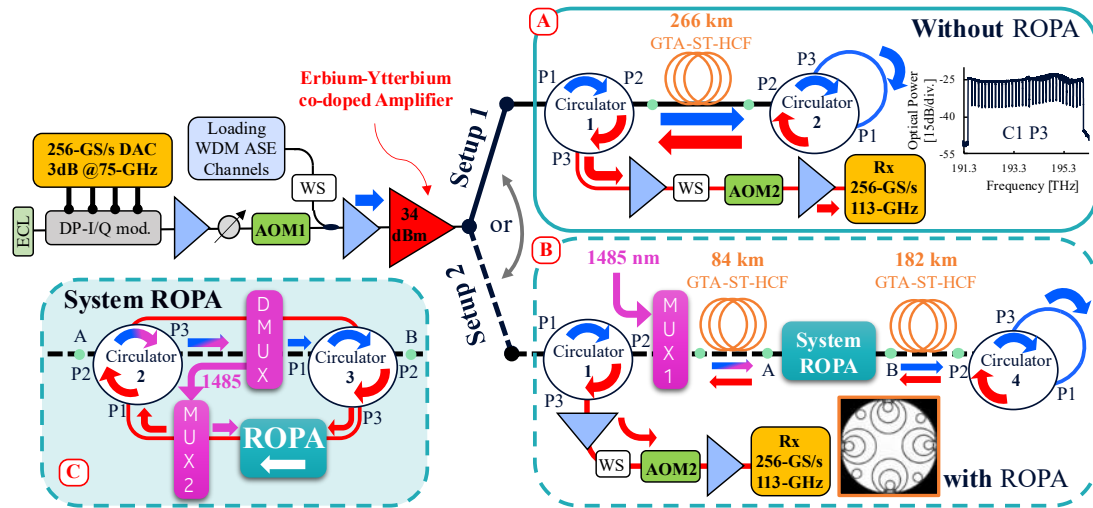


Fig. 3: Unrepeated setups at 532-km: (a) without: "Setup 1", solid line (b) with: "Setup 2", dashed line, remote optically pumped amplifiers (ROPA) amplification and 34 dBm Erbium-Ytterbium co-doped amplifier (HP- EYDFA). (c) System ROPA

spectrum with a channel spacing of 137.5 GHz. One notch is created by the WS to enable insertion of the channel under test (CUT), the power level of which is manually adjusted by means of a variable optical attenuator (VOA) to match the neighboring channels.

The CUT is generated from an external cavity laser source modulated by a dual-polarization (DP) I/Q modulator driven by an arbitrary waveform generator (AWG) operating at 256 GS/s to produce Nyquist pulse-shaped (0.01 roll-off) 135 GBd DP-QPSK for «Setup 1» and DP-16QAM for "Setup 2". The emitted spectrum is carefully optimized to account for the gas absorption lines at the high frequency end of the C band. We launched higher power at the high-frequency channels at the cost of lower power at low-frequency channels, which allowed us to achieve error free transmission over the complete C-band. The loading comb and the CUT are then multiplexed with an optical coupler and sent to the HP-EYDFA. The HP-EYDFA operates with an input power of -1 dBm and an output power of 34 dBm. Its maximum noise figure (NF) is 4.7 dB. For «Setup 1» without the ROPA, two optical circulators were utilized to enable light to travel through the 266 km HCF segment twice, effectively

achieving a total transmission distance of 532 km. The concatenated bi-directional OTDR traces are presented alongside the estimated loss profile, which is fitted using a linear estimation of -0.098 dB/km at 1550 nm for 266 km. However, when accounting for lumped losses caused by the optical circulators and HCF/SMF adaptors, the total span loss is measured at 57.2 dB (Fig.3(a)). At the receiver side, the CUT is extracted from the WDM comb with a WS configured as a bandpass filter. The CUT for the offline system is then received by a standard coherent receiver front-end and sampled at 256-GS/s using a 113-GHz bandwidth real-time oscilloscope. The standard DSP suite consists of chromatic dispersion (CD) compensation, complex 2x2 MIMO adaptive equalization, frequency offset and phase compensation, and LMS equalizer to mitigate for transmitter I/Q imperfections. Although the 2-circulator technique can be applied even over 532 km owing to the extremely low Rayleigh backscattering of HCF, the long span still incurs a loss of 57.2 dB. The associated return signal is then weaker than the discrete back-reflection from the SMF/HCF adapter (typically ~50 dB). To mitigate this crosstalk due to our bidirectional propagation configuration, we

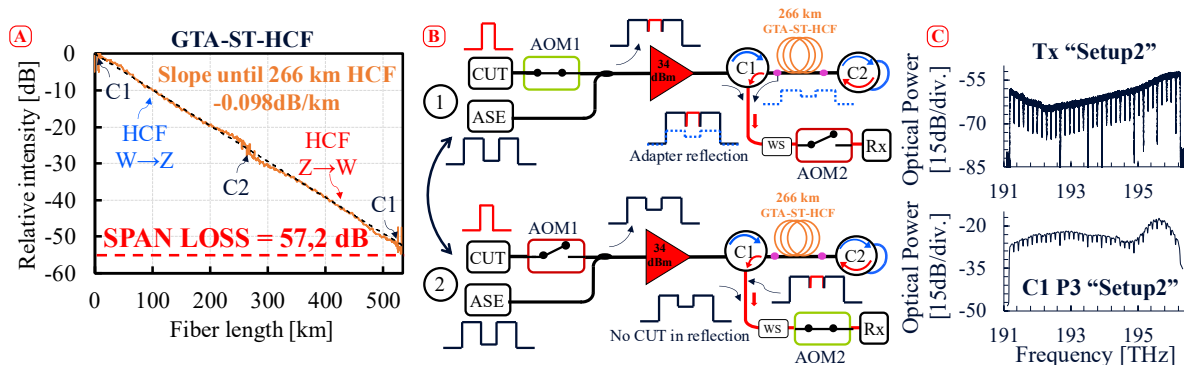


Fig. 2: "Setup 1" without ROPA (a) Concatenation OTDR measurement at 532 km with GTA-ST-HCF + 2 circulators, (b) Technique used to reduce crosstalk from the SMF/HCF free-space adaptor. (c) "Setup 2": Tx and Rx (C1P3) spectrum

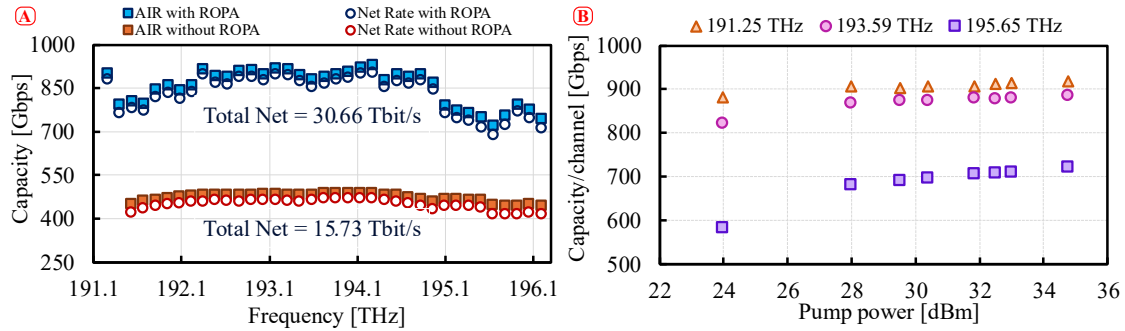


Fig. 4: (a) Transmission performance after 532-km with total net capacity of 15.73-Tbit/s in PDM-QPSK without ROPA and 30.66-Tbit/s in PDM-16QAM with ROPA. (b) Impact of the pump power on the signal capacity at three different frequencies.

designed a dynamic setup (shown in Fig. 3(b)) using two acousto-optic modulators (AOMs), that continuously switch between two configurations. Initially, in configuration (1), the CUT and ASE are launched into the HCF, while the AOM at the receiver is blocked to prevent adapter reflections. After a delay of approximately 1.7 milliseconds, once the signal propagates within the HCF, the system switches to configuration (2). In this state, the CUT transmission ceases (see the inset spectrum), and the signal that has propagated through 532 km is then detected by the receiver without the crosstalk from the SMF/HCF adaptor. For “Setup 2”, as we are leveraging the bi-directional propagation of the HCF, one cannot place the ROPA in the path without impacting the forward signal. We devised a ROPA system which would only amplify the return signal using 2 additional circulators, together with pump/signal demultiplexer and multiplexer as shown in the inset of Fig. 2(c). This system allows us to simultaneously perform bi-directional propagation while using ROPA. The ROPA, which simply consists of a piece of erbium doped fiber [18] is pumped using a laser typically at 1480 nm. In our case, we chose a pump at 1485 nm to leverage the low loss window of our GTA-ST-HCF, whose propagation losses have been measured to be ~ 0.13 dB/km. We positioned the ROPA at 84 km from the receiver end with the pump suffering approximately 15 dB losses including the multiplexers and the circulators. The power of the signal arriving at the ROPA is ~ -15 dBm. The ROPA pump power varied between 24-34.77 dBm at the fiber end. Such a configuration is representative of a ROPA system, because there is no non-linear interaction between the pump and the signal over the HCF. In Fig. 3(c), we implemented a dual-slope equalization at the transmitter (Tx), before HP-EYDFA, to optimize the total capacity and to reduce the impact of gas absorption at the upper end of Band C. We show the receiver (Rx) spectrum after circulator C1 in P3.

Experimental results

Fig.4. shows the measurements of the channels after 532 km (with and without ROPA). Each measurement consists of 5 recorded waveforms

of 8 million samples. Finally, we independently measure, on the demodulated signal, the generalized mutual information (GMI) as well as the net bit rate by applying a family of SC-LDPC codes with rates varying between 0.4 and 0.9 to achieve error-free decoding [19]. Fig. 4(a) shows the per channel achievable information rate (AIR) and net data rate at 135 Gbaud DP-QPSK with 35 channels in the C-Band without the use of ROPA amplification. The net data rate per channel ranges from 415 to 470 Gbit/s, despite the presence of gas absorption lines. We achieved a record total net capacity of 15.73-Tb/s after 532 km with a single booster amplifier. We then measured the per channel AIR and net data rate at 135 Gbaud DP-16QAM with 37 channels in the C-Band with “Setup 2” with the ROPA. The net data rate per channel ranges from 687 to 903 Gbit/s. We achieved a record total net capacity of 30.66-Tbit/s after 532 km. Unlike «Setup 1», the gas-line absorption has a more severe impact here because of the higher-order DP-16QAM format. In Fig.4(b), we show the impact of the pump power on the signal capacity at three different frequencies 191.25, 193.59 and 195.65 THz. By decreasing the pump power by a factor of 10, we only lose about 10.15 % capacity on average showing that one could operate at higher energy efficiency without significantly impacting the total capacity.

Conclusions

For the first time, we demonstrated complete C-band unrepeated WDM transmission systems over 532 km using a high-power EYDFA booster, achieving record net throughputs of 30.66-Tb/s with ROPA and 15.73-Tb/s without ROPA. These findings highlight hollow-core fiber as a promising technology for enabling next-generation ultra-long-reach unrepeated links, by leveraging low loss of the HCF for both signal propagation and remote pumping. This result improves the state of the art by more than 120 km in reach (30,7% increase). These results highlight hollow-core fiber and ROPA as key enablers for ultra-long-reach links, paving the way for future terrestrial, subsea, and remote-connectivity applications.

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